was so idealized, so devoid of spin, lumps, shocks, and radiation, that it should not be taken seriously.

Such skepticism held sway throughout the United States and Western Europe, but not in the U.S.S.R. There Lev Landau, still recuperating from his year in prison, kept a “Golden List” of the most important physics research articles published anywhere in the world. Upon reading the Oppenheimer–Snyder paper, Landau entered it in his List, and he proclaimed to his friends and associates that these latest Oppenheimer revelations had to be right, even though they were extremely difficult for the human mind to comprehend. So great was Landau’s influence that his view took hold among leading Soviet theoretical physicists from that day forward.

## Nuclear Interlude

Were Oppenheimer and Snyder right, or were they wrong? The answer would likely have been learned definitively during the 1940s had World War 11 and then crash programs to develop the hydrogen bomb not intervened. But the war and the bomb did intervene, and research on impractical, esoteric issues like black holes became frozen in time as physicists turned their full energies to weapons design.

Only in the late 1950s did the weapons efforts wind down enough to bring stellar implosion back into physicists’ consciousness. Only then did the skeptics launch their first serious attack on the Oppenheimer–Snyder predictions. Carrying the banner of the skeptics at first, but not for long, was John Archibald Wheeler. From the outset, a leader of the believers was Wheeler’s Soviet counterpart, Yakov Borisovich Zel’-dovich.

The characters of Wheeler and Zel’dovich were shaped in the fire of nuclear weapons projects during the nearly two decades that black-hole research was frozen in time, the decades of the 1940s and 1950s. From their weapons work, Wheeler and Zel’dovich emerged with crucial tools for analyzing black holes: powerful computational techniques, a deep understanding of the laws of physics, and interactive research styles in which they would continually stimulate younger colleagues. They also emerged carrying difficult baggage—a set of complex relationships with